Which Visual Elements Enable AI Geolocation?

Blur tests on 3,500 trials across five AI models

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Research summary accompanying the SCCUR 2026 poster

Abstract

Vision-language models demonstrate remarkable capabilities in applications such as captioning and visual question answering. However, complex machine learning models memorise more than the required task features. For instance, VLMs can help an adversary to infer where a photograph was taken, raising privacy concerns for shared images. This study investigates which visual elements VLMs rely on for geolocation. One hundred photographs with known locations, each in seven versions (original; Gaussian blur applied to text, buildings, nature, vehicles, or people; full-image blur), were submitted to five publicly accessible AI models (Gemini 3.6 Flash, Claude Sonnet 4.6, ChatGPT 5.5, Copilot, Meta AI), yielding 3,500 trials. Geolocation accuracy, the proportion of trials identifying the correct location, fell from 69.4% to 36.0% (text blurred), 46.4% (buildings), 58.2% (nature), 62.6% (vehicles), and 67.0% (people; not significant), versus 22.6% for full blur. Text blur significantly degraded all five models and building blur four of five (paired McNemar tests, Holm-corrected). To avoid querying models, pixel coverage was tested as a predictor of element importance. It agreed with measured importance better than chance (Kendall τ_b = 0.178) but underperformed a fixed importance ranking (τ_b = 0.505), which a 10-fold cross-validated per-category weighting nearly matched (τ_b = 0.491). Raw footprint is thus a poor proxy, as text is informative far beyond its area.

1 Introduction

Vision-language models can describe and answer questions about images. They can also help an adversary infer where a photograph was taken, which raises privacy concerns for shared images [1, 2].

Photo geolocation predates these models. PlaNet treated it as classification over cells of the globe; when well-traveled people played GeoGuessr against it and were asked what they relied on, they listed signs first, then vegetation, architectural style, lane-marking color, and the side of the road traffic drives on [3]. Theiner et al. measured how much each of 150 labeled object types shaped a convolutional network's geolocation guesses: towers, buildings, bridges, and mountains mattered for precise guesses, while grass, roads, water, and cars had very limited influence [4]. That model was not built to read text.

Question. Which visual elements do current AI chat assistants rely on to work out a photo's location? **Approach.** Blur one kind of content at a time and measure how much accuracy drops.

2 Methods

Design

Each of 100 photographs with a known ground-truth location was presented in seven versions: the unmodified original; five versions in which one category of content (text, buildings, nature, vehicles, or people) was blurred with a Gaussian blur; and a version in which the whole image was blurred. Five AI models (Gemini 3.6 Flash, Claude Sonnet 4.6, ChatGPT 5.5, Copilot, and Meta AI Muse Spark) each received every version separately, with one fixed prompt, giving $100 \times 7 \times 5 = 3{,}500$ trials.

The design follows occlusion analysis, in which part of an image is hidden and the change in the model's output is measured [5]. Here whole categories of content are hidden instead of sliding squares, and accuracy is measured instead of the model's confidence. Blur is an established way to remove information from part of an image, and it works on models whose internals cannot be inspected [6], such as commercial chat assistants.

Exact prompt

> *"Please determine the precise location where this image was captured. Include a detailed explanation of the reasoning that leads to your conclusion. Conclude your response with a single line stating the exact location."*

Scoring

Each response was scored by hand against the ground truth as Correct, Incorrect, or Unable (the model declined or could not commit to a location). "Unable" is counted as incorrect unless stated otherwise.

Statistics

- **Each blur vs. the original, per model.** Every version of an image is answered by the same model, so outcomes are paired; the exact McNemar test is applied to the images whose outcome changed. Changes are reported in percentage points (pp) with image-level bootstrap 95% confidence intervals (10,000 resamples); accuracy intervals are Wilson intervals.
- **Multiple comparisons.** The main correction is Holm over all $5 \times 6 = 30$ tests. Holm within each model (6 tests) is a sensitivity check.
- **Pooled analysis.** Logistic regression fitted by GEE (exchangeable correlation, clustered by image) with blur condition and model as factors gives pooled odds ratios. A Wald test of the condition $\times$ model interaction (24 df) tests whether models differ in which elements they depend on.
- **Measured category importance.** For each image and category, the drop is the average over the five models of (original correct − blurred correct), ranging from -1 to 1.

3 Results

Main findings

1. **Text matters most.** Pooled accuracy fell by 33.4 pp with text blurred, the largest drop for any single category, and text blur produced a significant drop for all five models (Fig. 1).
2. **The pooled ordering is Text > Buildings > Nature > Vehicles > People** (Fig. 3, Table 1). Text and buildings are the two largest single-category drops for every model; the order of the smaller effects varies by model.
3. **Buildings** produced a significant drop for four of five models (all but Copilot, where it is significant only under the within-model correction).
4. **Nature and vehicles** have smaller effects that depend on the correction used. Nature is significant after the 30-test Holm correction for Claude and Meta AI, and for Gemini and ChatGPT under the within-model correction. Vehicles is significant when pooled ($p = 0.0001$) but in no single-model test after the 30-test correction.
5. **People** blur had no significant effect for any model (raw $p \geq 0.09$).
6. **Models differ** in how much they depend on each element (interaction $\chi^2(24) = 42.3$, $p = 0.012$; Fig. 2). Meta AI lost 25 points to both text and building blur, while ChatGPT lost 45 points to text and 25 to buildings.

Refusals

Refusals were concentrated in Copilot (Table 3): it answered "Unable" on 31 original images and on 66 with text blurred, 49 with buildings blurred, and 76 with the whole image blurred, compared with at most 7 for any other model in any condition. The increases for text and whole-image blur are significant (Holm $p < 0.0001$). Counting only the trials Copilot answered, its accuracy still fell from 55.1% to 23.5% with text blurred, so blurring text both increased refusals and made the remaining guesses less accurate.

Which categories matter for each image

A fixed ranking (Text > Buildings > Nature > Vehicles > People, computed on the other 99 images) agrees with each image's own ranking at a mean Kendall τ_b of 0.505 (Table 4). Blurring its top one or two categories captures 75% and 85% of the largest achievable accuracy drop, compared with 37% and 45% for a random choice.

Does a category's size predict its importance?

Pixel coverage (how much of the image a category fills) was measured with a pretrained detector for people and vehicles and with heuristic detectors for text, nature, and buildings, and ranking categories by coverage was compared with each image's measured ranking (Table 5). Pixel coverage beats chance (permutation $p < 0.001$) but falls well short of the fixed ranking: it captures 35% of the largest achievable drop with one category, versus 75%. Learning one weight per category closes most of the gap. A category's size is a poor guide to how much location information it carries, which fits text's large effect despite its small share of the image.

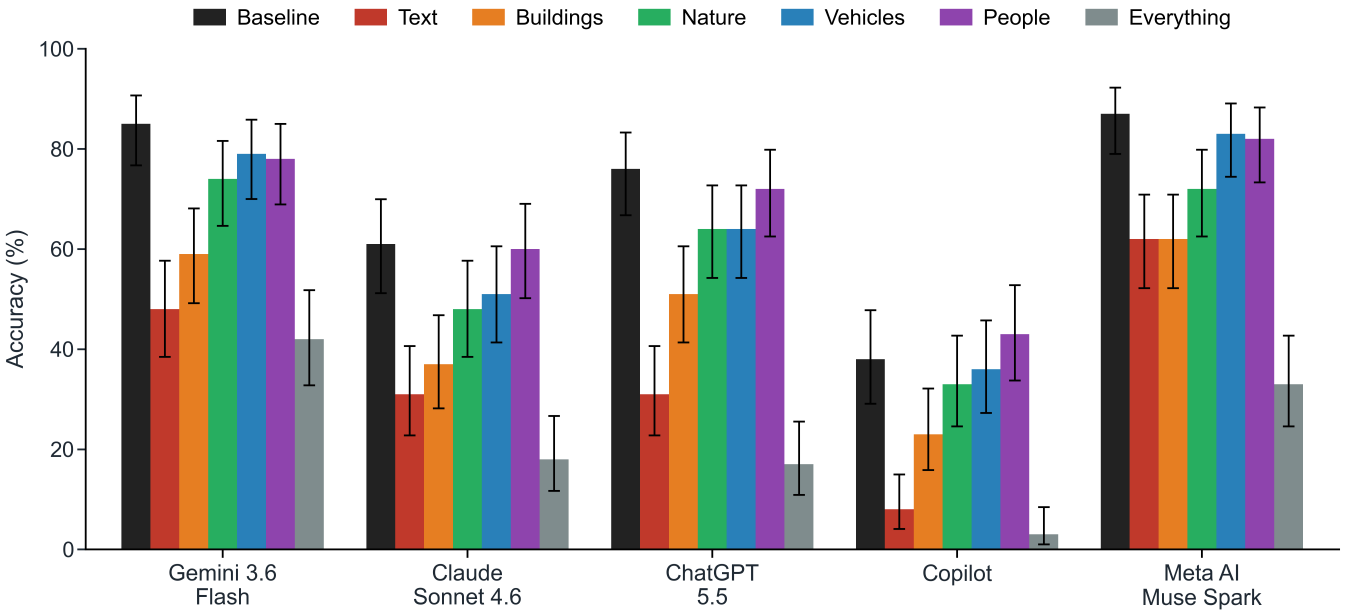


Figure 1: Accuracy by model and condition. Error bars: 95% Wilson confidence intervals. "Unable" counted as incorrect.

Table 1: Pooled effect of each blur across the five models. Change: mean over models with image-level bootstrap 95% CI. Odds ratio: logistic GEE with image-clustered errors, adjusted for model, original as reference.

Blurred	Pooled accuracy	Change, pp [95% CI]	Odds ratio [95% CI]	GEE p
None (original)	69.4%	—	—	—
Text	36.0%	−33.4 [−39.0, −28.0]	0.21 [0.15, 0.28]	< 0.0001
Buildings	46.4%	−23.0 [−28.6, −17.4]	0.33 [0.25, 0.44]	< 0.0001
Nature	58.2%	−11.2 [−15.0, −7.6]	0.57 [0.48, 0.69]	< 0.0001
Vehicles	62.6%	−6.8 [−10.2, −3.4]	0.71 [0.59, 0.84]	0.0001
People	67.0%	−2.4 [−5.6, +0.8]	0.88 [0.74, 1.05]	0.146
Everything	22.6%	−46.8 [−52.4, −41.0]	0.10 [0.07, 0.14]	< 0.0001



Figure 2: Change in accuracy versus the original image (pp). $*$ = significant after Holm correction over 30 tests.

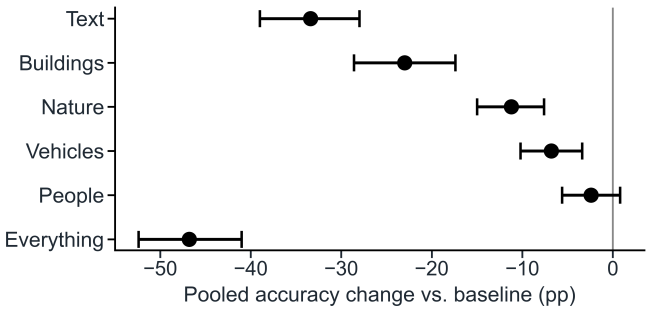


Figure 3: Pooled accuracy change versus the original (pp) with 95% bootstrap confidence intervals.

Table 2: Accuracy and paired tests versus the original image, per model (100 images each). Significant = 30-test Holm $p < 0.05$ (in bold). Under the within-model correction, nature for Gemini and ChatGPT, vehicles for Claude and ChatGPT, and buildings for Copilot are also significant.

Blurred	Accuracy [95% CI]	Change, pp [95% CI]	McNemar p	Holm p (30)	Significant
Gemini 3.6 Flash (baseline 85%)					
Text	48% [38, 58]	−37.0 [−47.0, −27.0]	< 0.0001	< 0.0001	**Yes**
Buildings	59% [49, 68]	−26.0 [−35.0, −17.0]	< 0.0001	< 0.0001	**Yes**
Nature	74% [65, 82]	−11.0 [−18.0, −4.0]	0.0074	0.0886	No
Vehicles	79% [70, 86]	−6.0 [−12.0, −1.0]	0.0703	0.6328	No
People	78% [69, 85]	−7.0 [−14.0, +0.0]	0.0923	0.7383	No
Everything	42% [33, 52]	−43.0 [−53.0, −33.0]	< 0.0001	< 0.0001	**Yes**
Claude Sonnet 4.6 (baseline 61%)					
Text	31% [23, 41]	−30.0 [−40.0, −20.0]	< 0.0001	< 0.0001	**Yes**
Buildings	37% [28, 47]	−24.0 [−33.0, −16.0]	< 0.0001	< 0.0001	**Yes**
Nature	48% [38, 58]	−13.0 [−20.0, −7.0]	0.0002	0.0039	**Yes**
Vehicles	51% [41, 61]	−10.0 [−17.0, −4.0]	0.0063	0.0829	No
People	60% [50, 69]	−1.0 [−8.0, +6.0]	1.0000	1.0000	No
Everything	18% [12, 27]	−43.0 [−53.0, −33.0]	< 0.0001	< 0.0001	**Yes**
ChatGPT 5.5 (baseline 76%)					
Text	31% [23, 41]	−45.0 [−55.0, −35.0]	< 0.0001	< 0.0001	**Yes**
Buildings	51% [41, 61]	−25.0 [−34.0, −16.0]	< 0.0001	< 0.0001	**Yes**
Nature	64% [54, 73]	−12.0 [−21.0, −4.0]	0.0118	0.1182	No
Vehicles	64% [54, 73]	−12.0 [−20.0, −4.0]	0.0075	0.0886	No
People	72% [63, 80]	−4.0 [−11.0, +3.0]	0.4240	1.0000	No
Everything	17% [11, 26]	−59.0 [−69.0, −49.0]	< 0.0001	< 0.0001	**Yes**
Copilot (baseline 38%)					
Text	8% [4, 15]	−30.0 [−39.0, −21.0]	< 0.0001	< 0.0001	**Yes**
Buildings	23% [16, 32]	−15.0 [−25.0, −5.0]	0.0059	0.0829	No
Nature	33% [25, 43]	−5.0 [−14.0, +4.0]	0.3593	1.0000	No
Vehicles	36% [27, 46]	−2.0 [−9.0, +5.0]	0.7744	1.0000	No
People	43% [34, 53]	+5.0 [−2.0, +12.0]	0.2668	1.0000	No
Everything	3% [1, 8]	−35.0 [−45.0, −26.0]	< 0.0001	< 0.0001	**Yes**
Meta AI, Muse Spark (baseline 87%)					
Text	62% [52, 71]	−25.0 [−35.0, −16.0]	< 0.0001	< 0.0001	**Yes**
Buildings	62% [52, 71]	−25.0 [−35.0, −16.0]	< 0.0001	< 0.0001	**Yes**
Nature	72% [63, 80]	−15.0 [−23.0, −7.0]	0.0007	0.0109	**Yes**
Vehicles	83% [74, 89]	−4.0 [−11.0, +3.0]	0.3877	1.0000	No
People	82% [73, 88]	−5.0 [−12.0, +2.0]	0.3018	1.0000	No
Everything	33% [25, 43]	−54.0 [−64.0, −43.0]	< 0.0001	< 0.0001	**Yes**

Table 3: "Unable" responses (out of 100 images) by model and condition.

Model	Original	Text	Buildings	Nature	Vehicles	People	Everything
Gemini 3.6 Flash	0	1	3	0	0	0	1
Claude Sonnet 4.6	0	0	0	0	0	0	1
ChatGPT 5.5	1	1	0	0	0	0	7
Copilot	31	66	49	34	34	26	76
Meta AI (Muse Spark)	1	1	1	1	0	1	0

Table 4: Measured category importance (100 images). Mean drop: per-image average over the five models of (original correct − blurred correct). Top share: fraction of images (with any measurable drop) where the category had the largest drop, ties included.

Category	Mean drop	Top share
Text	0.334	63%
Buildings	0.230	47%
Nature	0.112	18%
Vehicles	0.068	10%
People	0.024	7%

Table 5: Agreement between predicted and measured category importance: mean Kendall τ_b over 91 images where τ_b is defined (95% bootstrap CI), and share of the largest achievable drop captured by blurring the top 1 or 2 predicted categories.

Predictor	Mean τ_b [95% CI]	Top 1	Top 2
Pixel coverage	0.178 [0.105, 0.251]	35.3%	45.6%
Fixed ranking (leave-one-out)	0.505 [0.440, 0.567]	75.3%	85.5%
Learned per-category weights (10-fold CV)	0.491 [0.425, 0.553]	70.8%	85.5%
Random choice	0.000	36.7%	44.6%

4 Limitations

- 100 photographs, not a random sample of all images.
- One fixed prompt and one response per trial.
- Responses were scored by one person.
- Chat apps are not fully controlled: settings and model versions change over time.
- Blurring one element can change the rest of the image, and perturbations can introduce artifacts that affect a model's answer on their own [6]; some element detectors were heuristic.

References

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